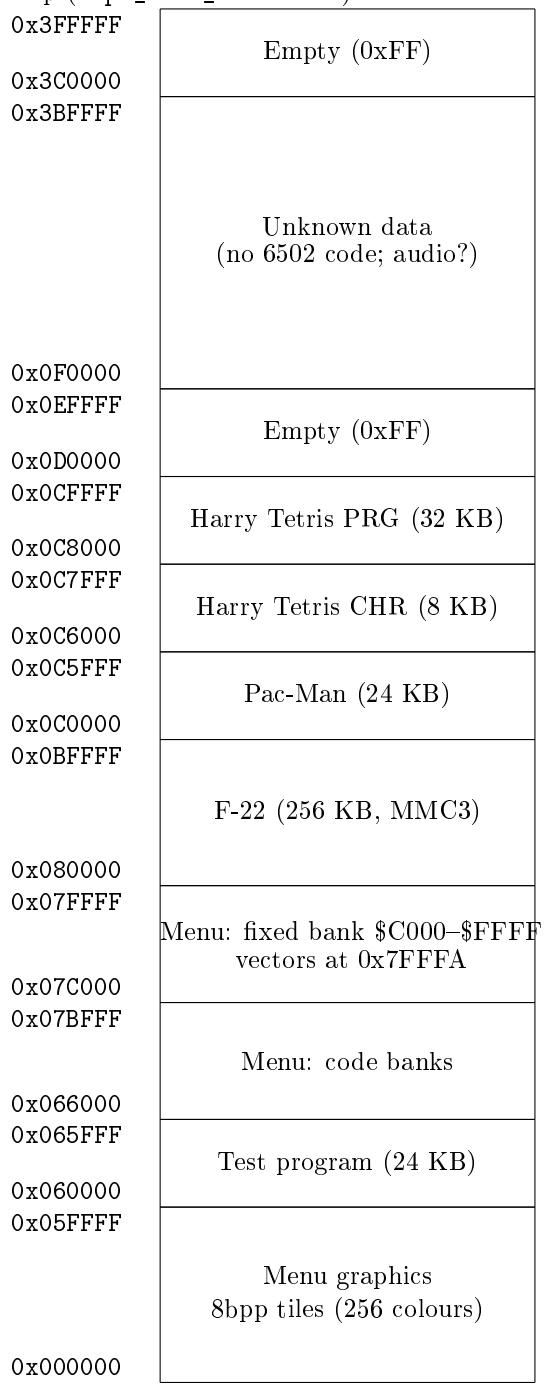
VPlay20K – W25Q32 flash map (4 MB)

Main chip: VT369 (OneBus, NES 2.0 mapper 256, console type 0x0A). Emulatable ROM: 16-byte NES 2.0 header + full dump (`vape_full_vt369.nes`).



Menu graphics (0x000000–0x05FFFF)

8×8 tiles in 8bpp: 1 byte per pixel, 64 bytes per tile. One block per 32 KB slice, padded with 0xFF:

Start	Used size
0x000000	0xF840
0x010000	0x6340
0x018000	0x5100
0x020000	0x5100
0x028000	0x3840
0x030000	0x3EC0
0x038000	0x2640
0x040000	0xA700
0x050000	0x4780
0x058000	0x5100

Menu code (fixed bank, CPU addresses)

Address	Role
$EEFB / $D520 / $D6AA	RESET / NMI / IRQ
$F113	Game launcher (banks $4107–$410A, jump to RAM)
$DD07 + table $DB08	LCD init, 160×128, 16-bit colour
$E189–$E35A	Bit-banged I2C ($4150/$4152, read $4153), device 0x40
$CC11 + table $E372	I2C challenge-response (anti-clone?), "Wellminds Technology"
$D3A9 / $D44C	BEAST / REGULAR confirmation
$EF74	Reads MCU status at boot
$D0CC	Power-down ($4175 = $19)

Link to the vape microcontroller (port $41EA/$41EB)

Direction	Bits	Meaning
VT369 → MCU	bit 6	Mode: 0 = BEAST (25 W), 1 = REGULAR
VT369 → MCU	bit 7	Unknown ($0D27)
MCU → VT369	bits 5–4 = 01	Battery, 0–5 ($0D30)
MCU → VT369	bits 5–4 = 11	E-liquid, 0–10 ($0D2F)
MCU → VT369	bit 5 at boot	Device status; bit 3 = saved mode

The 25 W, 3.5 V and 0.5 Ω values on the BEAST screen are fixed images: the actual heating power is set by the microcontroller, not by the VT369.